

Semantic Theory 2026: Exercise 3 Key

Question 1

Translate the following into λ -expressions. Use subscripts to indicate the types of the λ -bound variables (e.g. $\lambda x_e.P(x)$ for an e -type x).

- a. $pink_{\langle\langle e,t \rangle, \langle e,t \rangle\rangle}$ (as in “Jumbo is a pink elephant”; the expression should have $pink^*_{\langle e,t \rangle}$ as the underlying first-order predicate)

$$pink_{\langle\langle e,t \rangle, \langle e,t \rangle\rangle} = \lambda P_{\langle e,t \rangle} \lambda x_e [pink^*(x) \wedge P(x)]$$

- b. $and_{\langle e, \langle e, \langle\langle e,t \rangle, t \rangle \rangle \rangle}$ (as in “John and Suzy danced”; the expression should incorporate $\wedge$ as the underlying operator)

$$and_{\langle e, \langle e, \langle\langle e,t \rangle, t \rangle \rangle \rangle} = \lambda x_e \lambda y_e \lambda P_{\langle e,t \rangle} [P(x) \wedge P(y)]$$

- c. $not_{\langle\langle e,t \rangle, \langle e,t \rangle\rangle}$ (as in “Mark did not like the party”; the expression should incorporate $\neg$ as the underlying operator)

$$not_{\langle\langle e,t \rangle, \langle e,t \rangle\rangle} = \lambda P_{\langle e,t \rangle} \lambda x_e [\neg P(x)]$$

Question 2

Translate the following sentences into λ -expressions, assuming the syntactic structure indicated by the brackets. Then use lambda conversions (β -/ η -/ α -conversion) to reduce to λ -free terms.

Use the terms you derived for $pink$, and , and not in Question 1. If you weren’t able to derive those terms, you can simply use the predicates $pink'$, and' , and not' (with the types indicated in Question 1).

Ignore the contribution of past/plural morphology, “is”/“are”/“did”, and “a”.

- a. Jumbo [is a [pink elephant]]

$$\begin{aligned} & pink(elephant)(jumbo) \\ &= \lambda P_{\langle e,t \rangle} \lambda x_e [pink^*(x) \wedge P(x)](elephant)(jumbo) \\ &\Leftrightarrow_{\beta} \lambda x_e [pink^*(x) \wedge elephant(x)](jumbo) \\ &\Leftrightarrow_{\beta} pink^*(jumbo) \wedge elephant(jumbo) \end{aligned}$$

- b. [John and Suzy] danced

$$\begin{aligned} & and(suzy)(john)(dance) \\ &= \lambda x_e \lambda y_e \lambda P_{\langle e,t \rangle} [P(x) \wedge P(y)](suzy)(john)(dance) \\ &\Leftrightarrow_{\beta} \lambda y_e \lambda P_{\langle e,t \rangle} [P(suzy) \wedge P(y)](john)(dance) \\ &\Leftrightarrow_{\beta} \lambda P_{\langle e,t \rangle} [P(suzy) \wedge P(john)](dance) \\ &\Leftrightarrow_{\beta} dance(john) \wedge dance(john) \end{aligned}$$

c. Mark did [not [like the party]]

$$\begin{aligned}
& not(like(the-party))(mark) \\
& = \lambda P_{\langle e,t \rangle} \lambda x_e [\neg P(x)](like(the-party))(mark) \\
& \Leftrightarrow_{\beta} \lambda x_e [\neg like(the-party)(x)](mark) \\
& \Leftrightarrow_{\beta} \neg like(the-party)(mark)
\end{aligned}$$

d. [Tim [and Mary]] are [not [pink elephants]]

$$\begin{aligned}
& and(m')(t')(not(pink(elephant))) \\
& = \lambda x_e \lambda y_e \lambda P_{\langle e,t \rangle} [P(x) \wedge P(y)](m')(t')(\lambda R_{\langle e,t \rangle} \lambda z_e [\neg R(z)](\lambda Q_{\langle e,t \rangle} \lambda w_e [pink^*(w) \wedge Q(w)](elephant))) \\
& \Leftrightarrow_{\beta} \lambda x_e \lambda y_e \lambda P_{\langle e,t \rangle} [P(x) \wedge P(y)](m')(t')(\lambda R_{\langle e,t \rangle} \lambda z_e [\neg R(z)](\lambda w_e [pink^*(w) \wedge elephant(w)])) \\
& \Leftrightarrow_{\beta} \lambda x_e \lambda y_e \lambda P_{\langle e,t \rangle} [P(x) \wedge P(y)](m')(t')(\lambda z_e [\neg \lambda w_e [pink^*(w) \wedge elephant(w)](z)]) \\
& \Leftrightarrow_{\beta} \lambda x_e \lambda y_e \lambda P_{\langle e,t \rangle} [P(x) \wedge P(y)](m')(t')(\lambda z_e [\neg (pink^*(z) \wedge elephant(z))]) \\
& \Leftrightarrow_{\beta} \lambda y_e \lambda P_{\langle e,t \rangle} [P(m') \wedge P(y)](t')(\lambda z_e [\neg (pink^*(z) \wedge elephant(z))]) \\
& \Leftrightarrow_{\beta} \lambda P_{\langle e,t \rangle} [P(m') \wedge P(t')](\lambda z_e [\neg (pink^*(z) \wedge elephant(z))]) \\
& \Leftrightarrow_{\beta, \alpha} \lambda z_e [\neg (pink^*(z) \wedge elephant(z))](m') \wedge \lambda u_e [\neg (pink^*(u) \wedge elephant(u))](t') \\
& \Leftrightarrow_{\beta} \neg (pink^*(m') \wedge elephant(m')) \wedge \lambda u_e [\neg (pink^*(u) \wedge elephant(u))](t') \\
& \Leftrightarrow_{\beta} \neg (pink^*(m') \wedge elephant(m')) \wedge \neg (pink^*(t') \wedge elephant(t'))
\end{aligned}$$